VIPP Report

DIAGNijmegen AbdomenMRUS Prostate Segmentation Evaluation

Prostate158 Test Data (19 Cases)

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Report

Introduction

In this report, I evaluated the `joeranbosma/picai_prostate_segmentation_processor` model from the repository `DIAGNijmegen/AbdomenMRUS-prostate-segmentation` for whole-gland prostate MRI segmentation on a 19-case Prostate158 test set stored locally in the `prostate158_test` folder. The evaluated cases were `001` through `019`.

The official model description states that this algorithm segments the **whole prostate gland** in biparametric MRI (bpMRI) using an ensemble of five nnU-Net models trained with Cross-Entropy plus Focal loss. The documented model input consists of three aligned MRI volumes: `T2W`, `ADC`, and `HBV`, and the output is a single whole-gland prostate segmentation. A description of the model can be found at https://grand-challenge.org/algorithms/prostate-segmentation/ and the source code can be found at https://github.com/DIAGNijmegen/AbdomenMRUS-prostate-segmentation.

The dataset used here is the public Prostate158 test release. A dataset description can be found at https://doi.org/10.1016/j.compbiomed.2022.105817 and the public data release can be found at https://doi.org/10.5281/zenodo.6592345.

The original Prostate158 files were provided as `.nii.gz` volumes containing `t2`, `adc`, `dwi`, and multiple labels. To satisfy the model interface, the dataset was reorganized into the required three-input folder structure using `.mha` files: `transverse-t2-prostate-mri`, `transverse-adc-prostate-mri`, and `transverse-hbv-prostate-mri`. Since Prostate158 provides `dwi.nii.gz` rather than an explicit `hbv.nii.gz` file, the provided `dwi` volume was mapped into the model's HBV input slot for inference.

The model predicts a **binary whole-gland mask**. The provided Prostate158 anatomy labels `t2_anatomy_reader1` and `t2_anatomy_reader2` are multi-class labels with values {`0,1,2`}, so they were binarized to whole-gland masks for fair evaluation against the model output. The original provided labels were preserved separately. Tumor labels (`t2_tumor_reader1`, `adc_tumor_reader1`, and `adc_tumor_reader2`) were not used for scoring because this model is a whole-gland segmentation model, not a tumor model.

Docker GPU passthrough was available on the workstation and GPU inference completed successfully. After inference, the predicted masks were verified to be binary and spatially aligned with the prepared T2 inputs. The final mean Dice score was **0.9174** against reader 1, **0.9027** against reader 2, and **0.9101** when averaged across both readers over all **19** test cases. Geometry validation passed for all evaluated prediction-versus-ground-truth pairs.

Model Input Requirements and Label Handling

This evaluation answers the practical questions concerning how this model should be used on the Prostate158 test set:

1. What input does the model require?

The model requires three aligned prostate MRI inputs: `T2W`, `ADC`, and `HBV`. It does not operate as a single-sequence model.

2. What was used from Prostate158?

For each case, the following image files were used: `t2.nii.gz`, `adc.nii.gz`, and `dwi.nii.gz`. The `dwi` file was placed into the model's HBV input slot during preparation.

3. Which provided labels were relevant?

Only `t2_anatomy_reader1` and `t2_anatomy_reader2` were relevant for whole-gland evaluation. These were converted from multi-class zonal labels to binary whole-gland masks. Tumor labels were preserved but not used in Dice scoring.

4. What was the model output?

The model output was one binary whole-gland mask per case. The predictions were stored as `.mha` masks and also exported as `.seg.nrrd` files for direct review in 3D Slicer.

5. Were the review files altered?

The review segmentations for 3D Slicer were generated in `.seg.nrrd` format without changing the voxel arrays or image geometry of the masks. The conversion was verified by checking array equality and matching size, spacing, origin, and direction.

Qualitative Comparison

The following screenshots visually compare the two provided whole-gland references and the model prediction. In each figure, Reader 1 is shown in green, Reader 2 is shown in yellow, and the model prediction is shown in red.

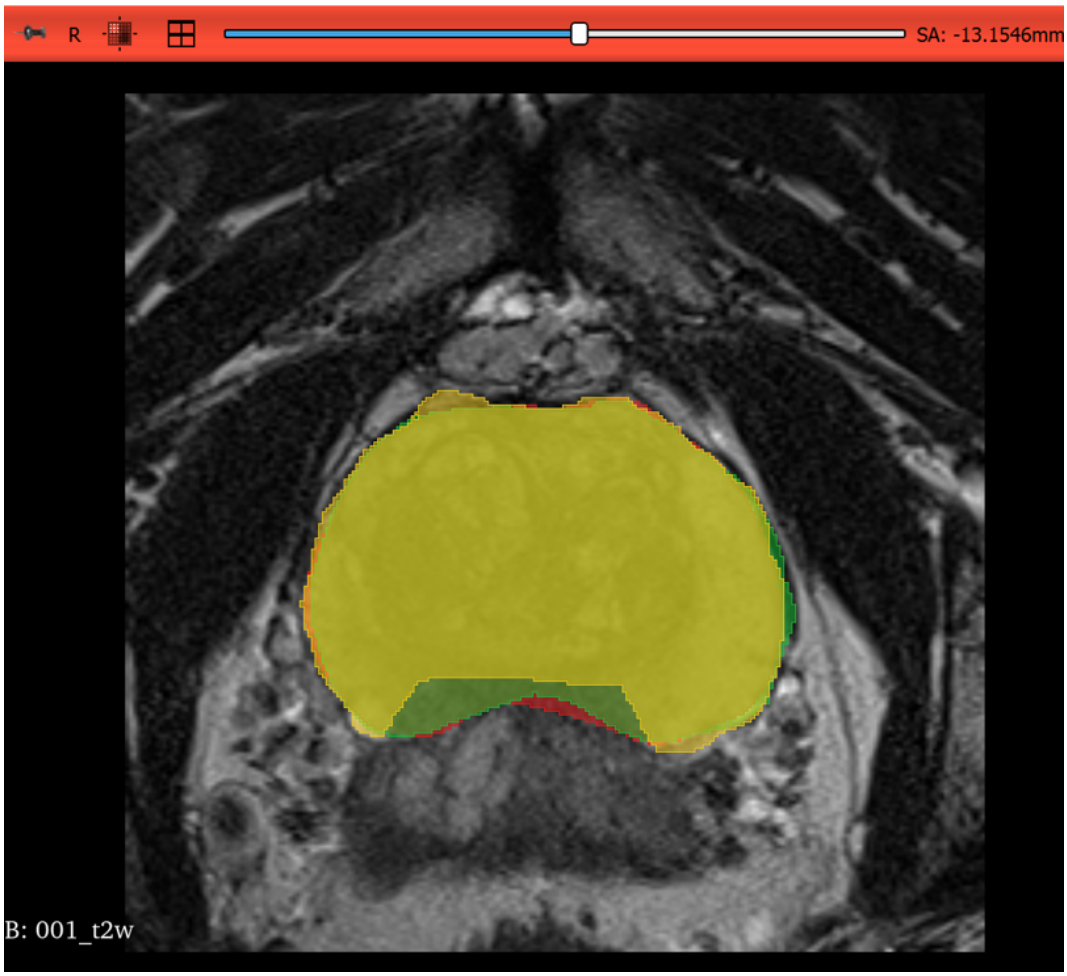

Figure 1: Qualitative comparison for Case001. Reader 1 in green, Reader 2 in yellow, and model prediction in red. Source image file: `case1.png`.

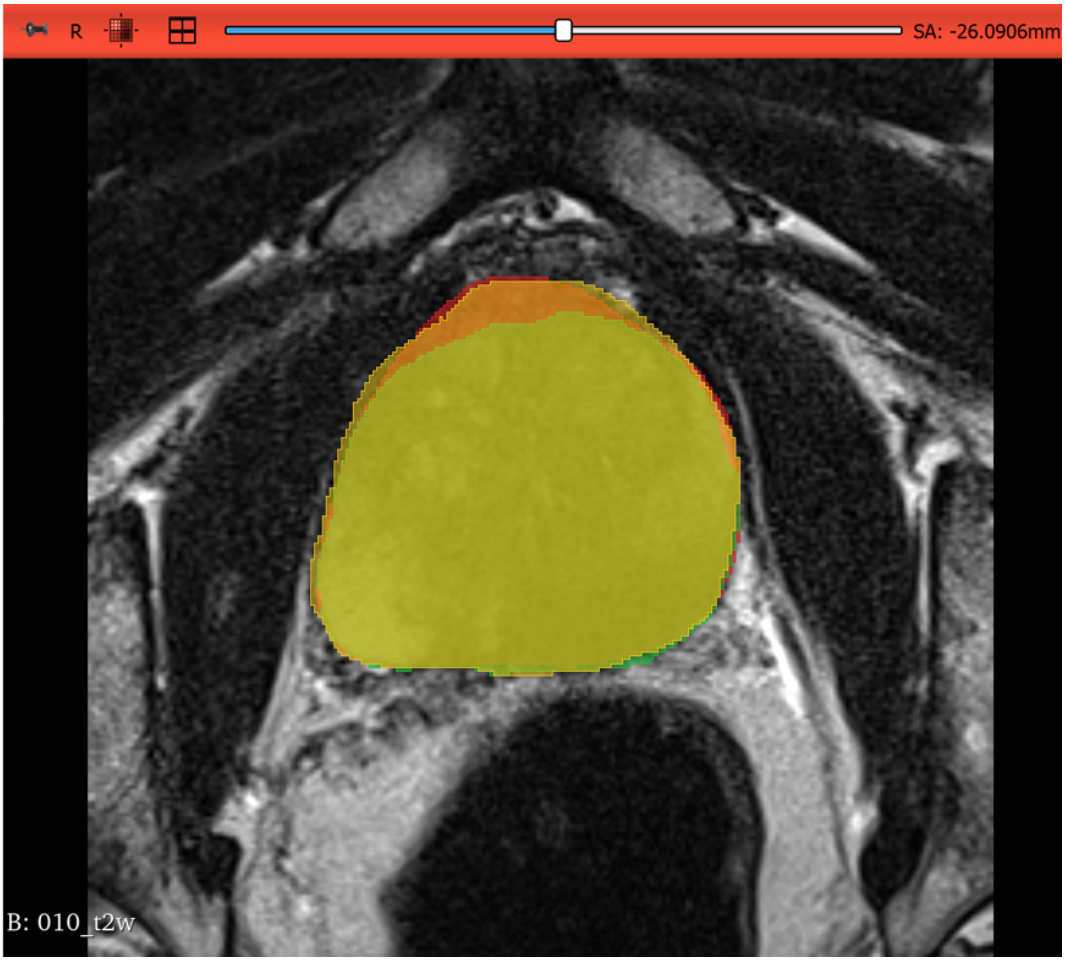

Figure 2: Qualitative comparison for Case010. Reader 1 in green, Reader 2 in yellow, and model prediction in red. Source image file: `case10.png`.

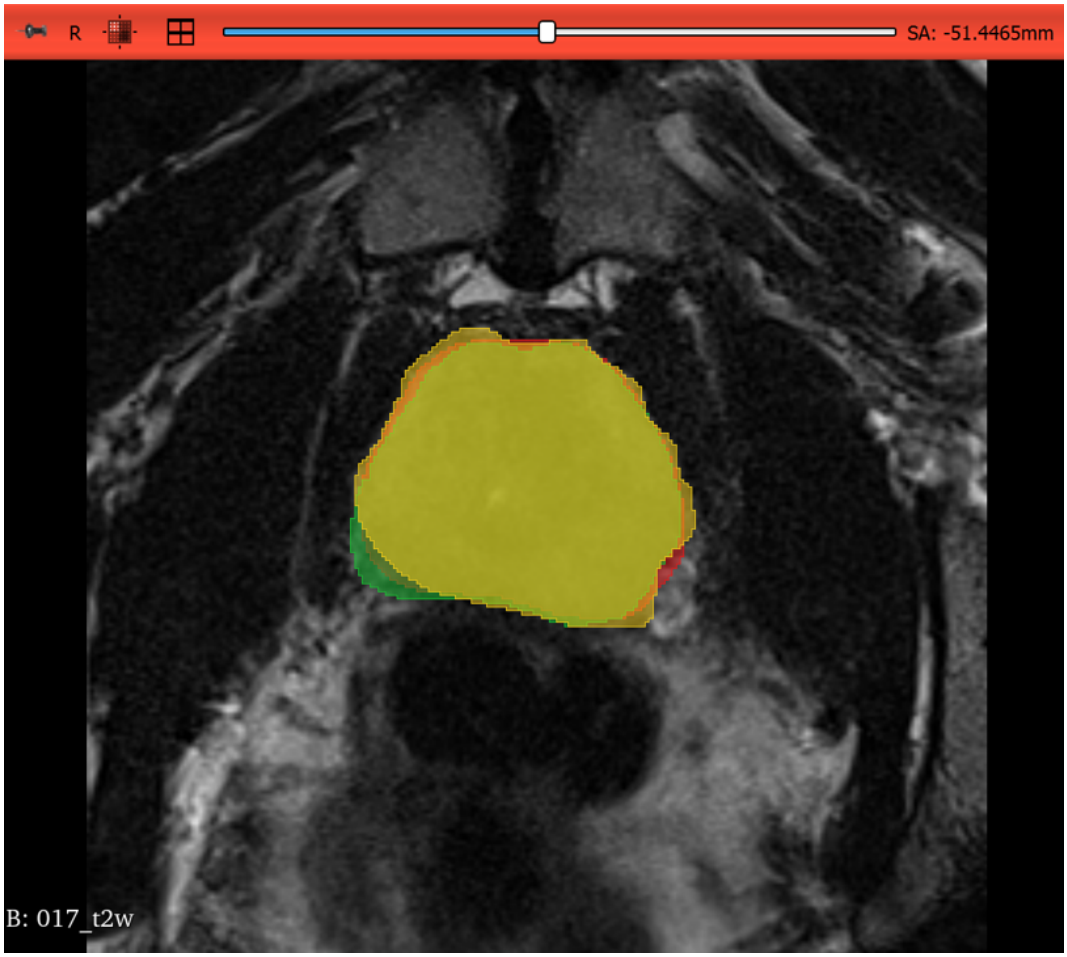

Figure 3: Qualitative comparison for Case017. Reader 1 in green, Reader 2 in yellow, and model prediction in red. Source image file: `case17.png`.

Quantitative Results

The Dice scores below compare the binary whole-gland model prediction against the binarized whole-gland anatomy masks from the two provided Prostate158 readers. Because the model predicts only the whole gland, zonal and tumor Dice scores are not applicable here.

Table 1: Dice score results obtained on the 19-case Prostate158 test evaluation set. Reader 1 and reader 2 refer to the two provided anatomy annotations after binarization to whole-gland masks.

Case	Task	Dice vs Reader 1	Dice vs Reader 2	Mean Dice
001	whole-gland	0.8878	0.8629	0.8754
002	whole-gland	0.9364	0.9116	0.9240
003	whole-gland	0.9346	0.9247	0.9297
004	whole-gland	0.9117	0.9077	0.9097
005	whole-gland	0.9387	0.9298	0.9342
006	whole-gland	0.9375	0.9377	0.9376
007	whole-gland	0.9198	0.9080	0.9139

Case	Task	Dice vs Reader 1	Dice vs Reader 2	Mean Dice
008	whole-gland	0.9222	0.8930	0.9076
009	whole-gland	0.9164	0.8700	0.8932
010	whole-gland	0.8833	0.8957	0.8895
011	whole-gland	0.9341	0.9274	0.9307
012	whole-gland	0.9282	0.9257	0.9270
013	whole-gland	0.8976	0.9070	0.9023
014	whole-gland	0.9250	0.9009	0.9130
015	whole-gland	0.9139	0.9257	0.9198
016	whole-gland	0.9199	0.8276	0.8737
017	whole-gland	0.9280	0.8849	0.9064
018	whole-gland	0.8700	0.8807	0.8754
019	whole-gland	0.9260	0.9303	0.9282
Dataset mean	whole-gland	0.9174	0.9027	0.9101
Dataset std	whole-gland	0.0191	0.0274	0.0200

Applicability to Other MRI Sequences

Based on the official model documentation, this model is intended for **biparametric prostate MRI examinations** with three aligned inputs: `T2W`, `ADC`, and `HBV`. It is not documented as a general-purpose prostate segmentation model for arbitrary MRI sequences, and it is not described as valid for `T1`-weighted MRI or for single-sequence-only inference.

The model documentation further states that the algorithm was developed for reasonably aligned bpMRI, that it was trained on Siemens scanner data with surface coils, and that it was not designed for cases with prior treatment, severe artefacts, or major sequence misalignment. Therefore, the answer is effectively **no** for supported use on non-bpMRI settings such as isolated T2-only, isolated ADC-only, isolated DWI-only, T1-weighted scans, or unrelated MRI protocols.

In this evaluation, the public Prostate158 data provided `dwi.nii.gz` rather than an explicitly named `hbv.nii.gz`. The DWI image was mapped into the model's HBV input slot for local inference because the model interface requires a third diffusion-derived input in that role. This local mapping worked operationally for the present dataset, but it should not be interpreted as proof that the model is validated for arbitrary diffusion variants outside its documented use case.

Relevant sources: Grand Challenge model page, GitHub repository, Prostate158 dataset paper.

Discussion

Overall, the model achieved strong whole-gland overlap on this 19-case Prostate158 test set. The best mean Dice score was obtained on `Case006` with **0.9376**. Other strong cases included `Case005` (0.9342), `Case011` (0.9307), `Case003` (0.9297), and `Case019` (0.9282). The weakest

mean result was `Case016` with **0.8737**, followed by `Case001` (0.8754) and `Case018` (0.8754). Even the weaker cases remained well above a Dice score of 0.85 except for the reader 2 comparison of `Case016`.

The two-reader evaluation is important because the Prostate158 anatomy labels were provided by two different annotators. In most cases, the model remained close to both references, but some cases showed notable inter-reader sensitivity. For example, `Case016` scored 0.9199 against reader 1 but 0.8276 against reader 2, indicating either annotation variability or a boundary-placement difference that materially affected overlap. This is one reason why the mean-over-both-readers score of **0.9101** is the most appropriate single-number summary for this experiment.

The official model page reports a cross-validation Dice similarity coefficient of 0.8968 on the developers' training and validation data. The present mean Dice of 0.9101 on the 19-case public Prostate158 test release is therefore consistent with strong whole-gland performance on same-domain bpMRI data. However, this should still be interpreted as an in-domain benchmark rather than proof of generalization to unrelated acquisition settings or unsupported MRI protocols.

In summary, this model performed well for binary whole-gland prostate segmentation on the Prostate158 test set when evaluated against binarized anatomy annotations from both readers. The reported Dice scores reflect whole-gland performance only; they do not evaluate zonal segmentation quality or tumor segmentation quality.